

Reloc3r: Large-Scale Training of Relative Camera Pose Regression for Generalizable, Fast, and Accurate Visual Localization

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Abstract

Visual localization aims to determine the camera pose of a query image relative to a database of posed images. In recent years, deep neural networks that directly regress camera poses have gained popularity due to their fast inference capabilities. However, existing methods struggle to either generalize well to new scenes or provide accurate camera pose estimates. To address these issues, we present **Reloc3r**, a simple yet effective visual localization framework. It consists of an elegantly designed relative pose regression network, and a minimalist motion averaging module for absolute pose estimation. Trained on approximately eight million posed image pairs, Reloc3r achieves surprisingly good performance and generalization ability. We conduct extensive experiments on six public datasets, consistently demonstrating the effectiveness and efficiency of the proposed method. It provides high-quality camera pose estimates in real time and generalizes to novel scenes. Code: <https://github.com/ffrivera0/reloc3r>.

1. Introduction

Visual localization, also known as camera re-localization, is a key challenge in computer vision, robotics, and graphics. It’s crucial for many applications, including augmented reality and robot navigation. The process involves registering new query images to a database of posed images or 3D models. This is typically done by estimating 6-degree-of-freedom (6-DoF) camera poses within a world coordinate system defined by the database.

Traditional visual localization approaches rely on structure-from-motion (SfM) techniques [27, 80, 88] to reconstruct 3D models. These methods [33, 37, 60, 72, 76, 80, 84, 95, 109, 112] match pixels from query images to 3D scene points, then use geometric optimization [36, 38, 47, 79, 81] to solve camera poses. While renowned for their

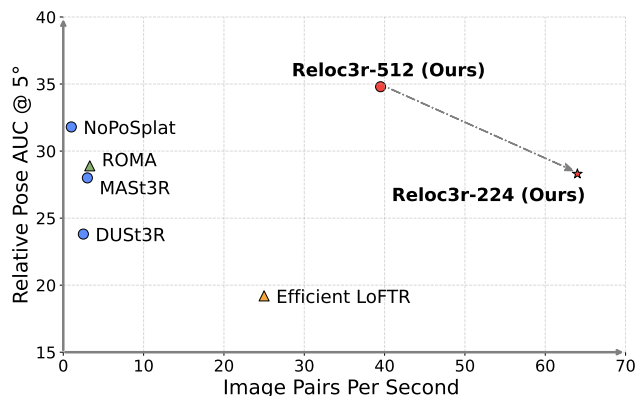


Figure 1. Comparison of pose accuracy and runtime efficiency. We report the AUC@5 and image pairs per second (FPS) on the ScanNet1500 [26, 80] dataset. We provide two versions of Reloc3r: one trained and tested on image widths of 512, and another on 224. The proposed Reloc3r-512 outperforms all other methods, achieving the best AUC@5 while maintaining an efficiency of 24 FPS. Remarkably, even at 224 resolution, our method matches ROMA [34] in accuracy while being 20× faster.

high localization accuracy, these approaches often struggle with inefficiencies at test time, which limits their scalability for real-time applications. Scene coordinate regression approaches [13–16, 30, 53, 54] provide an alternative perspective on pixel-to-point correspondences. These methods use neural networks to learn implicit scene representations, which are then used to infer dense correspondences. However, most of them face limitations in generalization. Moreover, these methods often require intensive supervision, such as ground-truth keypoint matches or 3D point maps, making it challenging to scale up the training data.

Absolute pose regression (APR) approaches [18, 42, 43, 66, 70, 89, 90, 103, 104, 117, 124] directly regress camera poses from images, offering much faster inference times and high accuracy. However, most of these methods are inherently scene-specific and typically require dense view-point coverage during training, limiting their real-world applicability. Recent attempts [20–22, 58, 69] to enhance ac-

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curacy through synthetic data generation introduce significant computational overhead, hampering widespread deployment. In contrast, relative pose regression (RPR) approaches [1, 4, 7, 35, 49, 49, 116, 129] estimate relative poses between database-query image pairs. These methods mitigate the need for per-scene training while maintaining the test-time efficiency of APR models. However, even the most advanced RPR methods [4, 29, 101, 116] have not yet matched APR approaches in localization accuracy. While several RPR methods [4, 49, 101, 116] have shown the ability to generalize across different datasets, this often results in further reduced camera pose accuracy. Therefore, most of the aforementioned approaches struggle with one of the following three criteria: novel-scene generalization, test-time efficiency, and camera pose accuracy.

To address these challenges, we present **Reloc3r** (pronounced “rɪˈləʊkər”), a simple yet surprisingly effective visual localization framework. It draws inspiration from recent foundation models [6, 46, 51, 63, 73, 98, 108, 111]. These models, utilizing scalable network architectures (such as Transformers [32, 40]) and large-scale training, demonstrate strong performance and exceptional generalization across various tasks. This success motivates our exploration of a similar methodology for pose estimation. We adopt architecture from DUST3R [111] as the backbone, applying parsimonious and elegant modifications to build a relative pose regression network. The network is designed to be fully symmetric and disregards the metric scale of relative poses during training. We then integrate this with a minimalist motion averaging module to estimate absolute poses, resulting in the Reloc3r framework. To unleash the power of large-scale training, we processed around eight million image pairs from diverse public sources, spanning object-centric, indoor, and outdoor scenes. Experimentally, Reloc3r demonstrates superior performance across six well-known pose estimation datasets, benefiting from its simple architecture and large-scale training. Our key contributions can be summarized as follows:

- We introduce Reloc3r, a simple yet surprisingly effective visual localization framework. It enables excellent generalization to novel scenes, fast test-time efficiency, and high camera pose accuracy.
- Both the proposed fully symmetric relative pose regression network and the motion averaging module follow the principle of parsimony. This streamlined approach enables efficient large-scale training.
- Comprehensive experiments across six popular evaluation datasets consistently demonstrate the effectiveness of our proposed approach.

2. Related Work

Structure-based visual localization. The structure-based localization pipeline is a well-established approach to solv-

ing camera pose via multi-view geometry [38]. Modern methods [37, 41, 55, 64, 72, 81–86, 96, 109, 120, 125, 129] typically consist of two main steps: 1) establishing correspondences either between images or between pixels and a pre-built 3D model, and 2) robustly solving the camera pose from noisy correspondences. These correspondences are obtained through keypoint matching [2, 27, 33, 34, 51, 60, 65, 76, 80, 95, 112] or scene coordinate regression [13–16, 30, 53, 54, 97, 110, 119]. Robust estimators [8–11, 25, 36, 48, 50] are then applied to estimate the final camera pose. Though effective, these methods often have slow inference times, which limits their use in real-time applications. Recent efficiency-oriented variants [60, 112] have been proposed to speed up the matching process. However, the complex system design and high computational cost of robust estimation remain bottlenecks. These methods typically require ground-truth correspondences or 3D point maps for supervision, limiting their scalability for large-scale training. To address these challenges, we choose a straightforward camera pose regression approach. This method allows for efficient large-scale training while simplifying the system, resulting in a faster and more scalable visual localization solution.

Camera pose regression. End-to-end pose regression [1, 4, 7, 18, 20–23, 29, 35, 42–44, 49, 58, 66, 69, 70, 78, 87, 89, 90, 101, 103, 104, 116, 117, 124, 129] has gained popularity due to its real-time inference capabilities. These methods fall into two broad categories: absolute pose regression (APR) and relative pose regression (RPR).

APR approaches directly regress camera positions and orientations in the world coordinate system from images within milliseconds. Despite their simplicity, these methods fall short of the localization accuracy achieved by structure-based approaches and often resemble pose approximation through image retrieval [87]. To improve the accuracy of APR, some methods [21, 22, 58, 69] resort to novel view synthesis [67] to create dense viewpoints for training. While being effective, this strategy introduces significant computational costs, with training taking hours or days for each specific scene. More recent approach [23] attempts to reduce training time to minutes by building the connection between pose regressor and scene-specific maps. However, it is still limited to per-scene training and evaluation.

RPR approaches aim to generalize across different scenes by learning the relative pose between image pairs. Localization is achieved by regressing the relative pose between the query image and the most similar (or top- K) database images. The metric scale of the relative translation can be estimated approximately from a single database-query pair [1, 4, 7, 35], while more precise absolute positioning is possible through multi-view triangulation [31, 49, 116, 129]. Despite this capability, the best RPR methods [4, 29, 45, 116, 129] lag significantly behind APR meth-

ods. Furthermore, the generalization ability of existing RPR models remains limited. For instance, methods like Relative PN [49] and Relpose-GNN [101] can adapt to new datasets and scenes, but their localization errors nearly double in these scenarios. Map-free [4], despite being trained on large datasets (about 523K samples), still relies on separate models for indoor and outdoor settings. Other approaches explore multi-view pose estimation [57, 106, 118, 127, 128], as well as wide-baseline [77] and panoramic methods [100]. However, they remain dataset-specific training, which limits their scalability.

Previous methods have largely focused on technical design rather than scaling training on larger and more diverse datasets. In this paper, we present the first pose regression approach trained on a diverse mixture of object-centric, indoor, and outdoor datasets. Contrary to prior work [129] suggesting that pose regression inaccuracies stem from coarse feature localization, we find that a well-trained patch-level regression network can achieve, and sometimes even surpass the performance of pixel-level feature matching.

Foundation Models. Large neural network models have advanced significantly due to the scalability of the Transformer architecture [102]. This architecture underpins both large language models [5, 24, 28, 68, 71, 99] and large vision models [6, 32, 46, 63, 73, 74, 98]. These foundation models, trained on large-scale datasets, have proven effective and demonstrate strong generalization capabilities across diverse tasks. Recently, DUST3R [111] introduces the first 3D foundation model capable of addressing nearly all tasks related to two-view geometry. Its architecture has been adapted and fine-tuned in numerous follow-up works [51, 92, 105, 122, 126] to enhance performance in various downstream tasks. In our work, we adopt DUST3R’s Transformer backbone to develop Reloc3r, but featuring a fully symmetric design. We will discuss the benefits of this symmetric architecture in the following section.

3. Method

Problem statement. Given a database $\mathbf{D} = \{I_{d_n} \in \mathbb{R}^{H \times W \times 3} \mid n = 1, \dots, N\}$ of posed images from a scene, and a query image I_q from the same scene, the task of visual localization is to estimate the 6-DoF camera pose $P \in \mathbb{R}^{3 \times 4}$ that can register I_q to the world coordinate system defined by the database images. P is represented by camera rotation $R \in \mathbb{R}^{3 \times 3}$ and translation $t \in \mathbb{R}^3$.

Method overview. Figure 2 illustrates an overview of the proposed visual localization method. It comprises two main components: a *relative pose regression network*, and a *motion averaging module*. The query image I_q is paired with the top- K database images using an off-the-shelf image retrieval approach [3], creating a set of image pairs $\mathbf{Q} = \{(I_{d_k}, I_q) \mid k = 1, \dots, K\}$. The relative pose re-

gression network processes each image pair from $\mathbf{Q}$ independently to determine their relative poses $P_{d,q}$ and $P_{q,d}$. With known poses of database images $\{\hat{P}d_1, \dots, \hat{P}d_K\}$, it is possible to infer the query image’s absolute rotation from a single pair. However, these estimates can be noisy, and the translation vector’s metric scale remains uncertain. To tackle these issues, a motion averaging module performs rotation averaging and camera center triangulation, eventually producing the absolute metric pose of the query image.

In Sec. 3.1, we provide a detailed description of the pose regression model. This model leverages a unified Vision Transformer (ViT) [32] architecture with minimal modifications. The motion averaging module discussed in Sec. 3.2 also takes a minimalist approach. It performs simple rotation averaging and camera center triangulation, without any trainable parameters involved.

3.1. Relative Camera Pose Regression

The proposed relative camera pose regression network, inspired by DUST3R [111], takes an image pair I_1, I_2 as input. While we assume the images have the same resolution (consistent with DUST3R), their resolutions may differ in practice. The network divides the images into patches and processes them as tokens through a Vision Transformer (ViT) encoder. A ViT decoder [113, 114] then uses a cross-attention mechanism to exchange information between tokens from both branches. This is followed by regression heads that predict the relative poses $\hat{P}_{I_1, I_2}$ and $\hat{P}_{I_2, I_1}$. The two branches are fully symmetrical with shared weights.

ViT encoder-decoder architecture. The two modules resemble those of DUST3R and gain substantial benefits from pre-training. We begin by dividing each input image I_i into a sequence of T tokens, each with dimension d . Next, we compute RoPE positional embeddings [94] for each token to encode their relative spatial positions within the image. Then, we process the tokens through m ViT encoder blocks, each comprising self-attention and feed-forward layers, to produce encoded feature tokens F_1 and F_2 :

$$F_i^{(T \times d)} = \text{Encoder}(\text{Patchify}(I_i^{(H \times W \times 3)})), \text{ where } i = 1, 2.$$

The decoder comprises n ViT decoder blocks, each using the same RoPE position embedding. Unlike the encoder block, each decoder block incorporates an additional cross-attention layer between its self-attention and feed-forward layers. This structure enables the model to reason the spatial relationship between two sets of feature tokens. We obtain decoded tokens as:

$$\begin{aligned} G_1^{(T \times d)} &= \text{Decoder}(F_1^{(T \times d)}, F_2^{(T \times d)}), \\ G_2^{(T \times d)} &= \text{Decoder}(F_2^{(T \times d)}, F_1^{(T \times d)}). \end{aligned}$$

Pose regression head. Following recent research [23], our pose regression head comprises h feed-forward layers followed by average pooling, with additional layers to regress

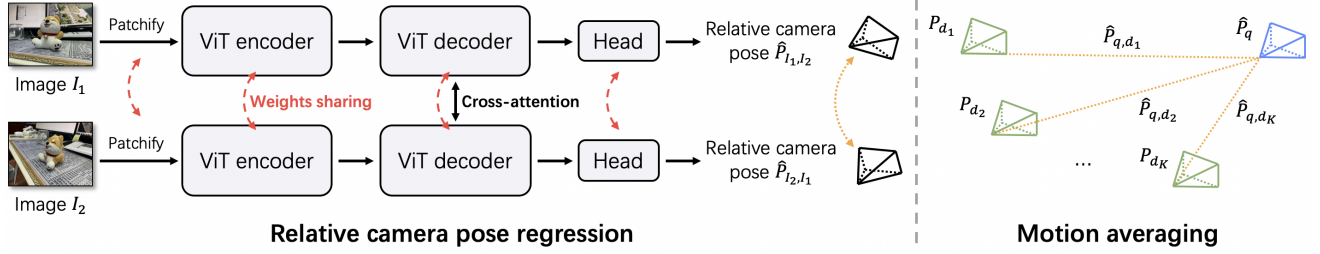


Figure 2. Reloc3r consists of two modules: a relative camera pose regression network (Sec. 3.1) and a motion averaging module (Sec. 3.2). Given a pair of input images, the network module infers the relative camera pose (at an unknown scale) between them. This module consists of a two-branch Vision Transformer (ViT) with shared weights. The images are divided into patches, converted to tokens, and embedded as latent features through separate encoders. Decoders then exchange information between the two sets of latent features. Each head aggregates its latent features to estimate a relative camera pose. To determine the absolute camera pose of a query image relative to a database, we retrieve at least two database-query pairs. These pairs are first processed by the network for relative pose estimation. Subsequently, the motion averaging module computes the absolute metric pose by aggregating the relative estimates.

relative rotation and translation. The rotation is initially represented using a 9D representation [52]. It is then converted into a 3×3 rotation matrix using SVD orthogonalization. This matrix is concatenated with the 3D translation vector to form the final transformation matrix. The final output for the relative pose is:

$$\hat{P}_{I_1, I_2}^{(3 \times 4)} = \text{Head}(G_1^{(T \times d)}), \quad \hat{P}_{I_2, I_1}^{(3 \times 4)} = \text{Head}(G_2^{(T \times d)}).$$

Supervision signal. The predicted pose by our network conveys two pieces of information: 1) rotation, which measures the relative change in orientation, and 2) translation, which indicates the relative direction of camera center movement. Both of these quantities can be expressed as relative angles. Consequently, we train the network by minimizing the difference between these predicted relative angles and their ground-truth: $\mathcal{L} = \ell_R + \ell_t$, where

$$\ell_R = \arccos\left(\frac{\text{tr}(\hat{R}^{-1}R) - 1}{2}\right), \ell_t = \arccos\left(\frac{\hat{t} \cdot t}{\|\hat{t}\| \|t\|}\right).$$

Here, $\text{tr}(\cdot)$ denotes the trace of a matrix. $\hat{R}$ and $\hat{t}$ represent the predicted rotation and translation, while R and t represent their respective ground-truth values.

Discussion. Unlike DUST3R’s asymmetric branches for coordinate alignment, we use a fully symmetric architecture, which is inherently more suitable for relative pose estimation. This design eliminates biases from image ordering, thereby simplifying training. It also allows weight sharing across branches, reducing computational complexity and storage requirements.

Recent works [4, 116] on relative pose regression prefer learning a metric pose. In contrast, we choose to learn only the direction of translation, as motion averaging can effectively solve the metric scale (discussed in the next section). This approach represents translations as angles in the same dimension as rotations, eliminating the need to weight ro-

tation and translation values during training. It also avoids the challenge of balancing different dataset metric scales.

3.2. Motion Averaging

To maintain efficiency and simplicity, we integrate our pose regression network with a minimalist motion averaging module. Given the high accuracy of the network’s predictions, we don’t apply any robust estimation [19, 31, 36, 115, 131]. For each image pair, the regression network produces two relative poses: $\hat{P}_{I_1, I_2}$ and $\hat{P}_{I_2, I_1}$, which ideally should be inverses of each other. Empirically, we observe similar accuracy for both poses. By default, we use the transformation that maps the query to the database as input for motion averaging. The module processes rotation and translation separately, as detailed below.

Rotation averaging. Given a relative rotation estimation $\hat{R}_{q, d_i}$ from a database-query pair, the absolute rotation is computed as $\hat{R}_q = R_{d_i} \hat{R}_{q, d_i}$. The motion averaging module reduces prediction noise by aggregating absolute rotation estimates from all available pairs. This aggregation is efficiently performed by calculating the mean rotation using quaternion representations [129]. We have observed that calculating the median rotation can further enhance robustness against noise, with minimal additional computational cost. Consequently, we use the median rotation as our final rotation estimation.

Camera center triangulation. The absolute camera center can be triangulated [39] from two database-query pairs. Similar to rotation averaging, we calculate an average intersection using all valid pairs. While the geometric median of the intersection is not analytically solvable and typically requires iterative optimization, we opt for a more efficient approach. We use a simple least-squares method that minimizes the sum of squared distances from the camera center to each translation direction derived from relative pose estimates. The solution is obtained through Singular Value

Decomposition (SVD).

4. Experiments

In this section, we first introduce our training data and implementation details, then evaluate our method across multiple datasets. In Sec. 4.1, we verify our network’s performance on the relative pose estimation task. In Sec. 4.2, we evaluate our full method on the visual localization task. Finally, we conduct ablation studies and analyses in Sec. 4.3. It’s worth noting that our method is trained once for all experiments, meaning all evaluations use the same model, except for the ablation studies.

Training data. To build up-to-scale training pairs with ground-truth relative poses, similar to DUST3R [111], we process around eight million image pairs from seven public datasets. The statistics are reported in Table 1. We convert the ground-truth relative poses to a unified format: relative coordinate transformation aligned with the OpenCV [17] definition. Each image is center-cropped based on its principal points and resized to a width of 512 pixels.

Datasets	Scene type	Number of image pairs
CO3Dv2 [75]	Object-centric	~1M
ScanNet++ [123]	Indoor	~850K
ARKitScenes [12]	Indoor	~2.1M
BlendedMVS [121]	Outdoor	~1M
MegaDepth [56]	Outdoor	~1.8M
DL3DV [61]	Indoor & outdoor	~1.1M
RealEstate10K [130]	Indoor & outdoor	~100K

Table 1. Training data of Reloc3r. It comprises about eight million image pairs from seven public sources, covering a wide range of scenes from object-centric to indoor and outdoor environments.

Implementation details. By default, the proposed Reloc3r uses $m = 24$ encoder blocks, $n = 12$ decoder blocks, followed by the pose regression head with $h = 2$ convolutional layers. To improve memory usage and speed in self and cross-attention, we employ memory-efficient attention. This design achieves approximately a 14% speed increase while saving 25% of GPU memory during training. We initialize Reloc3r with DUST3R’s pre-trained 512-DPT weights. For decoder initialization, we use the weights from DUST3R decoder2, as it’s pre-trained to perform coordinate transformation. The full model is trained on 8 AMD MI250x-40G GPUs with a batch size of 8 and a learning rate starting at $1e-5$, decaying to $1e-7$.

For the visual localization task, following the literature [79, 101], we apply NetVLAD [3] for image retrieval and use the top 10 similar image pairs. We directly use these retrieved image pairs without distance-based clustering, filtering, or other heuristics to enhance their spatial distribution. All evaluations are conducted on a 24GB NVIDIA GeForce RTX 4090 GPU and we use mixed precision fp16

	Methods	RRA@15	RTA@15	mAA@30
Non-PR	PixSfM [59]	33.7	32.9	30.1
	RelPose [127]	57.1	-	-
	PoseDiffusion [106]	80.5	79.8	66.5
	RelPose++ [57]	82.3	77.2	65.1
	RayDiffusion* [128]	93.3	-	-
	VGGSFm [107]	92.1	88.3	74.0
	DUST3R (w/ PnP) [111]	94.3	88.4	77.2
	MASt3R [51]	94.6	91.9	81.1
PR	PoseReg [106]	53.2	49.1	45.0
	RayReg* [128]	89.2	-	-
	Reloc3r-224 (Ours)	93.6	91.9	79.1
	Reloc3r-512 (Ours)	95.8	93.7	82.9

Table 2. Relative pose evaluation (multi-view) on the CO3Dv2 dataset [75]. The best results for each method category are highlighted in bold. The underlined numbers indicate that our method achieves the best performance among all competitors. The methods marked with * donate evaluation on 8 frames.

/ fp32 to improve speed and memory efficiency without sacrificing accuracy.

4.1. Relative Camera Pose Estimation

In this section, we evaluate Reloc3r’s relative pose regression module. Our evaluation covers two scenarios: pair-wise relative pose on ScanNet1500 [26, 80], RealEstate10K [130], and ACID [62] datasets, as well as multi-view relative pose on Co3dv2 [75] dataset.

Pair-wise relative pose. Following recent research [122], we evaluate pair-wise relative pose on three datasets: ScanNet1500 [26, 80], RealEstate10K [130], and ACID [62]. These datasets showcase a variety of indoor and outdoor scenes captured with diverse camera trajectories. ScanNet1500 [26, 80] focuses on indoor scenes, RealEstate10K [130] covers both indoor and outdoor environments, and ACID [62] is an aerial outdoor dataset. Importantly, our evaluation test sets do not overlap with scenes in our training data, and ACID remains entirely unseen by Reloc3r during training. Following previous research [80, 95, 112], we employ three metrics: AUC@5/10/20. These metrics calculate the area under the curve of pose accuracy using thresholds of $\tau = 5/10/20$ degrees for the minimum of rotation and translation angular errors.

Since no existing pose regression (PR) methods have been specifically designed for these datasets, we primarily compare our approach with non-PR methods. Additionally, we evaluate the current state-of-the-art relative pose regression methods, ExReNet [93, 116] and Map-free [4], on these datasets. The results are presented in Table 3. Reloc3r outperforms other PR methods by a significant margin across all three datasets. When compared to non-PR methods, Reloc3r also achieves SoTA performance. Specifically, on the ScanNet1500 dataset, Reloc3r delivers top

	Methods	ScanNet1500			RealEstate10K			ACID			Inference time
		AUC@5	AUC@10	AUC@20	AUC@5	AUC@10	AUC@20	AUC@5	AUC@10	AUC@20	
Non-PR	Efficient LoFTR [112]	19.20	37.00	53.60	-	-	-	-	-	-	40 ms
	ROMA [34]	28.90	50.40	68.30	54.60	69.80	79.70	46.30	58.80	68.90	300 ms
	DUST3R [111]	23.81	45.91	65.57	39.70	56.88	70.43	21.50	35.95	49.70	441 ms
	MASt3R [51]	28.01	50.24	68.83	63.54	76.39	84.50	52.12	64.54	73.61	294 ms
	NoPoSplat [122]	31.80	53.80	71.70	69.10	80.60	87.70	48.60	61.70	72.80	>2000 ms
PR	Map-free (Regress-SN) [4]	1.84	8.75	25.33	0.83	4.06	13.97	1.32	5.82	16.28	10 ms
	Map-free (Regress-MF) [4]	0.50	3.48	13.15	1.61	6.74	18.38	2.57	9.96	24.50	10 ms
	ExReNet (SN) [116]	2.30	10.71	26.13	2.17	7.94	20.43	1.90	7.53	18.69	17 ms
	ExReNet (SUNCG) [116]	1.61	7.00	18.03	3.27	12.06	27.85	4.14	13.43	27.70	17 ms
	Reloc3r-224 (Ours)	28.34	52.60	71.56	59.70	75.05	84.71	28.25	47.34	62.54	15 ms
	Reloc3r-512 (Ours)	<u>34.79</u>	<u>58.37</u>	<u>75.56</u>	66.70	80.20	88.39	38.18	56.39	70.34	25 ms

Table 3. Relative camera pose evaluation on the ScanNet1500 [26, 80], RealEstate10K [130] and ACID [62] datasets. The best results for each method category are highlighted in bold. Our method outperforms all the pose regression competitors. Moreover, it achieves the best results (underlined) among all the competitors on several datasets and metrics. Notably, our method runs in real-time, which is over 50× faster than the state-of-the-art Non-PR method. Please refer to Figure 1 for an intuitive comparison with representative methods.

	Methods	Chess	Fire	Heads	Office	Pumpkin	RedKitchen	Stairs	Average	Dataset-specific training time
APR	LENS [69]	0.03 / 1.30	0.10 / 3.70	0.07 / 5.80	0.07 / 1.90	0.08 / 2.20	0.09 / 2.20	0.14 / 3.60	0.08 / 3.00	Days / scene
	PMNet [58]	0.03 / 1.26	0.04 / 1.76	0.02 / 1.68	0.06 / 1.69	0.07 / 1.96	0.08 / 2.23	0.11 / 2.97	0.06 / 1.93	Days / scene
	DFNet [21]+NeFeS [22]	0.02 / 0.57	0.02 / 0.74	0.02 / 1.28	0.02 / 0.56	0.02 / 0.55	0.02 / 0.57	0.05 / 1.28	0.02 / 0.79	Days / scene
	Marepo [23]	0.02 / 1.24	0.02 / 1.39	0.02 / 2.03	0.03 / 1.26	0.04 / 1.48	0.04 / 1.71	0.06 / 1.67	0.03 / 1.54	15min / scene
RPR (Seen)	EssNet (7S) [129]	-	-	-	-	-	-	-	0.22 / 8.03	Hours
	Relative PN (7S) [49]	0.13 / 6.46	0.26 / 12.72	0.14 / 12.34	0.21 / 7.35	0.24 / 6.35	0.24 / 8.03	0.27 / 11.82	0.21 / 9.30	Hours
	NC-EssNet (7S) [129]	-	-	-	-	-	-	-	0.21 / 7.50	Hours
	RelocNet (7S) [7]	0.12 / 4.14	0.26 / 10.4	0.14 / 10.5	0.18 / 5.32	0.26 / 4.17	0.23 / 5.08	0.28 / 7.53	0.21 / 6.73	Hours
	Relpose-GNN [101]	0.08 / 2.70	0.21 / 7.50	0.13 / 8.70	0.15 / 4.10	0.15 / 3.50	0.19 / 3.70	0.22 / 6.50	0.16 / 5.20	Hours
	AnchorNet [78]	0.06 / 3.89	0.15 / 10.3	0.08 / 10.9	0.09 / 5.15	0.10 / 2.97	0.08 / 4.68	0.10 / 9.26	0.09 / 6.74	Hours
	CamNet [29]	0.04 / 1.73	0.03 / 1.74	0.05 / 1.98	0.04 / 1.62	0.04 / 1.64	0.04 / 1.63	0.04 / 1.51	0.04 / 1.69	Hours
RPR (Unseen)	EssNet (CL) [129]	-	-	-	-	-	-	-	0.57 / 80.06	None
	NC-EssNet (CL) [129]	-	-	-	-	-	-	-	0.48 / 32.97	None
	Relative PN (U) [49]	0.31 / 15.05	0.40 / 19.00	0.24 / 22.15	0.38 / 14.14	0.44 / 18.24	0.41 / 16.51	0.35 / 23.55	0.36 / 18.38	None
	RelocNet (SN) [7]	0.21 / 10.9	0.32 / 11.8	0.15 / 13.4	0.31 / 10.3	0.40 / 10.9	0.33 / 10.3	0.33 / 11.4	0.29 / 11.3	None
	ImageNet+NCM [129]†	-	-	-	-	-	-	-	0.19 / 4.30	None
	Map-free (Match) [4]†	0.10 / 2.93	0.12 / 4.95	0.11 / 5.40	0.12 / 3.01	0.16 / 3.19	0.14 / 3.45	0.21 / 4.50	0.14 / 3.92	None
	Map-free (Regress) [4]	0.09 / 2.66	0.13 / 4.54	0.11 / 4.81	0.11 / 2.77	0.16 / 3.11	0.14 / 3.48	0.18 / 4.70	0.13 / 3.72	None
	ExReNet (SN) [116]	0.06 / 2.15	0.09 / 3.20	0.04 / 3.30	0.07 / 2.17	0.11 / 2.65	0.09 / 2.57	0.33 / 7.34	0.11 / 3.34	None
	ExReNet (SUNCG) [116]	0.05 / 1.63	0.07 / 2.54	0.03 / 2.71	0.06 / 1.75	0.07 / 2.04	0.07 / 2.10	0.19 / 4.87	0.08 / 2.52	None
	Reloc3r-224 (Ours)	0.03 / 0.99	0.04 / 1.13	0.02 / 1.23	0.05 / 0.88	0.07 / 1.14	0.05 / 1.23	0.12 / 2.25	0.05 / 1.26	None
	Reloc3r-512 (Ours)	0.03 / 0.88	0.03 / 0.81	<u>0.01 / 0.95</u>	0.04 / 0.88	0.06 / 1.10	0.04 / 1.26	0.07 / <u>1.26</u>	0.04 / 1.02	None

Table 4. Visual localization results on the 7 Scenes dataset [91]. We report median pose errors in meters and degrees. The best results for each method category are highlighted in bold. The underlined numbers indicate where Reloc3r outperforms all competitors. The methods marked with † indicate hybrid pose estimation, which combines additional geometric solvers with feature matching.

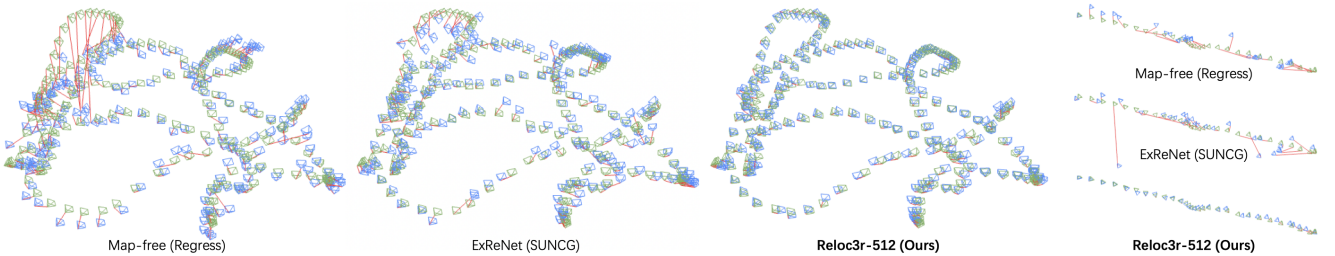


Figure 3. We visualize pose estimates for two scenes: Chess from the 7 Scenes dataset [91] and KingsCollege from Cambridge Landmarks [44]. We compare Reloc3r’s results with those of the most closely related RPR methods: ExReNet [116] and Map-free [4]. We can observe that Reloc3r’s pose estimates align more closely with the ground-truth poses.

	Methods	GreatCourt	KingsCollege	OldHospital	ShopFacade	StMarysChurch	Average (4)	Average	Dataset-specific training time
APR	LENS [69]	-	0.33 / 0.50	0.44 / 0.90	0.27 / 1.60	0.53 / 1.60	0.39 / 1.20	-	Days / scene
	PMNet [58]	-	0.31 / 0.55	0.44 / 0.79	0.17 / 0.86	0.31 / 0.96	0.31 / 0.79	-	Days / scene
	DFNet [21]+NeFeS [22]	-	0.37 / 0.54	0.52 / 0.88	0.15 / 0.53	0.37 / 1.14	0.35 / 0.77	-	Days / scene
RPR (Seen)	EssNet (CL) [129]	-	-	-	-	-	1.08 / 3.41	-	Hours
	Relpose-GNN [101]	3.20 / 2.20	0.48 / 1.00	1.14 / 2.50	0.48 / 2.50	1.52 / 3.20	0.91 / 2.30	1.37 / 2.30	Hours
	NC-EssNet (CL) [129]	-	-	-	-	-	0.85 / 2.82	-	Hours
	AnchorNet [78]	-	0.57 / 0.88	1.21 / 2.55	0.52 / 2.27	1.04 / 2.69	0.84 / 2.10	-	Hours
RPR (Unseen)	EssNet (7S) [129]	-	-	-	-	-	10.36 / 85.75	-	None
	NC-EssNet (7S) [129]	-	-	-	-	-	7.98 / 24.35	-	None
	Map-free (Match) [4]†	9.09 / 5.33	2.51 / 3.11	3.89 / 6.44	1.04 / 3.61	3.00 / 6.14	2.61 / 4.83	3.90 / 4.93	None
	Map-free (Regress) [4]	8.40 / 4.56	2.44 / 2.54	3.73 / 5.23	0.97 / 3.17	2.91 / 5.10	2.51 / 4.01	3.69 / 4.12	None
	ExReNet (SN) [116]	10.97 / 6.52	2.48 / 2.92	3.47 / 3.90	0.90 / 3.27	2.60 / 4.98	2.36 / 3.77	4.08 / 4.32	None
	ExReNet (SUNCG) [116]	9.79 / 4.46	2.33 / 2.48	3.54 / 3.49	0.72 / 2.41	2.30 / 3.72	2.22 / 3.03	3.74 / 3.31	None
	ImageNet+NCM [129]†	-	-	-	-	-	0.83 / 1.36	-	None
	Reloc3r-224 (Ours)	1.71 / 0.94	0.47 / 0.41	0.87 / 0.66	0.18 / 0.53	0.41 / 0.73	0.48 / 0.58	0.73 / 0.65	None
	Reloc3r-512 (Ours)	1.22 / 0.73	0.42 / 0.36	0.62 / 0.55	0.13 / 0.58	0.34 / 0.58	0.38 / 0.52	0.55 / 0.56	None

Table 5. Visual localization results on the Cambridge Landmarks [44]. We report median pose errors in meters and degrees. The best results for each method category are highlighted in bold. The underlined numbers indicate where Reloc3r outperforms all competitors. The methods marked with † indicate hybrid pose estimation, which combines additional geometric solvers with feature matching.

performance with AUC at all thresholds, outperforming its baseline DUST3R [111] by around 13% on AUC@20. On the RealEstate10K dataset, Reloc3r achieves performance comparable to NoPoSplat [122], with AUC values of 66.70 / 80.20 / 88.39% at thresholds of 5, 10, and 20, respectively. Similarly, on the ACID dataset, Reloc3r achieves high accuracy (AUC@20 = 70.34%), demonstrating its generalization ability to unseen datasets. Furthermore, Reloc3r runs at an inference time of just 42 ms at an image resolution of 512 in width. This is significantly faster than many non-PR methods such as NoPoSplat [122] (>2000 ms) and ROMA [34] (300 ms), and on par with PR methods. Such speed makes Reloc3r ideal for real-time applications.

Multi-view relative pose. We evaluate the multi-view relative pose on the Co3dv2 [75] dataset. This dataset consists of object-level scenes captured with inward-facing camera trajectories. The main challenges of this dataset include visual symmetries, textureless objects, and wide baselines between images. Following the evaluation protocol in [106, 111], we evaluate Reloc3r on the test sets of 41 categories. For each sequence, we randomly sample 10 frames and formulate all 45 pairs for the evaluation. These relative poses are evaluated with three metrics: the relative rotation accuracy within 15 degrees (RRA@15), the relative translation accuracy within 15 degrees (RTA@15), and the mean average accuracy (mAA@30, also called AUC@30).

We compare Reloc3r with recent SfM-based approaches, PixSfM [59] and VGGSfM [107], as well as data-driven methods including RelPose [127], RelPose++ [57], PoseDiffusion [106], RayDiffusion [128], DUST3R (with PnP) [111], and MAST3R [51]. In addition, we also compare the regression-based methods PoseReg [106] and RayReg [128]. Note that while we refer to our evaluation as multi-view, we actually only use pair-wise evaluation,

similar to DUST3R (w/ PnP) and MAST3R. In contrast, all other methods (excluding RayDiffusion and RayReg use 8 frames) use all 10 frames simultaneously, thus having more contextual information for evaluation.

The quantitative results are reported in Table 2. Compared to existing approaches, the proposed Reloc3r achieves SoTA performance across multiple metrics, demonstrating significant improvements over competing methods. Specifically, Reloc3r achieves the best scores in terms of RRA@15 (95.8%), RTA@15 (93.7%), and mAA@30 (82.9%), surpassing both Non-PR and PR methods. This highlights Reloc3r’s ability to robustly and accurately localize images in multi-view settings and wide baselines.

4.2. Visual Localization

In this section, we evaluate Reloc3r’s absolute pose estimation for visual localization. We conduct experiments using two public datasets: the 7 Scenes dataset [91] and Cambridge Landmarks [44]. The 7 Scenes dataset comprises seven indoor room scenes, each containing several video sequences captured from different moving trajectories. Cambridge Landmarks is a suburban-scale outdoor dataset featuring six scenes. Following prior approaches [15, 54, 101], we use five of these scenes for evaluation. For each scene, we report the median translation and rotation errors (in meters and degrees, respectively). It’s worth noting that both datasets were entirely unseen by Reloc3r during training.

Indoor visual localization. We compare Reloc3r with state-of-the-art absolute pose regression (APR) and relative pose regression (RPR) methods on the 7 Scenes [91] dataset. The results are shown in Table 4. For RPR methods, we categorize them into two groups: *seen*, where the model is trained and evaluated on the same dataset, and *unseen*, where the dataset is entirely new to the model during

Methods	ScanNet1500		
	AUC@5	AUC@10	AUC@20
Reloc3r-512 asymmetric	32.71	56.84	74.63
Reloc3r-512 metric pose	25.70	50.20	70.07
Reloc3r-512 (default)	34.79	58.37	75.56

Table 6. Ablation studies examining asymmetric network architecture and pose prediction with metric scales.

evaluation. The table shows that methods like EssNet [129], Relative PN [49], and RelocNet [7] experience significant performance drops when evaluated on unseen datasets, revealing their limitations in scene generalizability. In contrast, our method consistently outperforms all RPR methods, even those trained on the same dataset, achieving an average median error of $0.04m/1.02^\circ$. Moreover, when compared with APR methods, our approach demonstrates comparable performance without requiring scene-specific training, further highlighting Reloc3r’s robustness and adaptability across diverse scenes.

Outdoor visual localization. For the Cambridge [44] dataset, a widely used benchmark for regression-based localization, all RPR methods (both *seen* and *unseen*) face challenges in this outdoor setting. Our proposed Reloc3r, as shown in Table 5, surpasses all previous RPR methods without any retraining or fine-tuning on specific scenes, achieving consistent improvements across all evaluated scenes. Notably, in *unseen* conditions, Reloc3r demonstrates an average halving of the pose error compared to the previous state-of-the-art RPR method, ImageNet+NCM [129], with average errors of $0.38m/0.52^\circ$ across the last four scenes. Furthermore, our method shows a better average rotation error than all APR-based methods. This supports our motivation that leveraging a simple architectural design, coupled with scaled-up training, can effectively achieve good performance. Visual comparisons are shown in Figure 3.

4.3. Analyses

We summarize most insights here and refer the reader to the supplementary material for more details and discussions.

Image resolutions. We train and evaluate Reloc3r at two image resolutions: widths of 224 and 512 pixels. The results are shown in the tables in Sec. 4.1 and Sec. 4.2. Similar to DUST3R [111], the higher resolution improves accuracy but increases runtime due to processing more tokens.

Comparison with asymmetric network architecture. We train an asymmetric version of Reloc3r-512 that uses separate decoders and heads for each branch. As shown in Table 6, this version performs even worse than the default Reloc3r while requiring more computational resources.

Comparison with pose prediction with metric scale. We train a version of Reloc3r-512 with metric poses as output. The lower accuracy results reported in Table 6 validate the

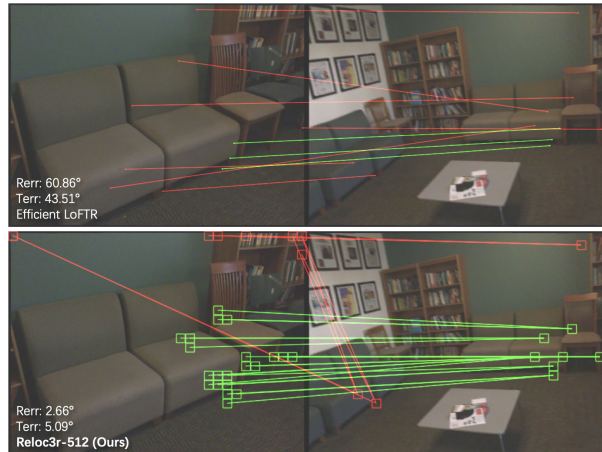


Figure 4. The top row showcases matches from Efficient LoFTR, while the bottom row displays the top-3 cross-attention responses from Reloc3r’s decoder. We observe that the correlated regions in Reloc3r are superior to those of Efficient LoFTR, even though Reloc3r is trained solely with pose supervision.

effectiveness of our default design without metric scales.

Interesting findings. We visualize the cross-attention responses in some blocks from Reloc3r’s decoder. Interestingly, we find that several layers have developed the ability to match patch correspondences, as illustrated in Figure 4.

Limitations. A failure case of Reloc3r occurs when the query image and all the retrieved database images are perfectly collinear, leading to a degeneracy issue that makes the metric scale unsolvable using motion averaging methods.

5. Conclusion

In this paper, we present Reloc3r, a simple yet effective visual localization framework. It consists of an elegantly designed relative pose regression network with a minimalist motion averaging module. Leveraging large-scale training on around eight million image pairs, Reloc3r demonstrates strong generalization capability, high efficiency, and accurate pose estimation performance across multiple datasets, while remaining parsimonious. We hope it will advance research on data scale and diversity in visual localization.

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